Chapter 5

The Internet & Its Uses

Cambridge IGCSE & O Level Computer Science

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Covers: 5.1 Internet & WWW - 5.2 Digital currency

5.3 Cyber security

Ready-to-learn notes with definitions, diagrams, exam tips & past-paper style practice

5.1 The Internet & the World Wide Web

5.1.1 Internet vs WWW – don't mix them up!

Exam trap – this is one of the MOST commonly confused pairs in the whole syllabus.

Internet	World Wide Web (WWW)
Worldwide collection of interconnected networks & devices	A part of the internet – collection of web pages accessed via a browser
Send/receive emails, online chat	Uses HTTP(S), written in HTML
Uses TCP & IP protocols	URLs specify page locations; accessed via web browsers

5.1.2 URLs (Uniform Resource Locators)

```
protocol :// website address / path / file name
https://www.hoddereducation.co.uk/ict
```

- Protocol – usually http or https
- Website address – domain host (www) + domain name + domain type (.com/.org/.gov) + optional country code (.uk/.pk)
- Path – the web page (often omitted, becomes root directory)
- File name – the specific item on the page

5.1.3 HTTP vs HTTPS

HTTP: rules for transferring files across the internet. Add SSL/TLS security, it becomes HTTPS ("s" = secure) – look for the green padlock.

5.1.4 Web Browser Features

Home page – bookmarks (favourites) – history – back/forward navigation – multiple tabs – cookies – hyperlinks – cache – JavaScript support – address bar

5.1.5 DNS - Finding a Web Page

> ***DNS (Domain Name Server):*** converts a human-readable URL into the numeric IP address a computer understands, e.g. www.hoddereducation.co.uk into 107.162.140.19

1. User types URL – browser asks DNS server (1) for the IP

↓

2. Not found locally – DNS server (1) asks DNS server (2)

↓

3. DNS server (2) finds the IP, sends it back to server (1) which caches it

↓

4. IP address sent back to user's computer

↓

5. Computer connects to the website server, page downloads

Fig 5.3 - how DNS locates a web page

5.1.6 Cookies

> ***Cookie:*** a small file/code sent by a web server to store on the user's browser as (key, data) pairs – e.g. (language, English).

Session Cookie	*Persistent Cookie*
No expiry date given, treated as session	Has an expiry date
Stored in temporary memory	Stored on the hard drive
Deleted when browser/session closes	Remains until expiry or manually deleted
Used for e.g. shopping basket items	Remembers login details, preferences

> ***Exam tip*** – persistent cookies are also called "tracking cookies" because they follow browsing habits across visits to target ads.

5.2 Digital Currency

Digital currency: money that exists purely in digital format (no physical form), unlike fiat currency ($, £, €).

Traditional digital currency relies on a central bank – Nick's Bank X – Central Bank – Irina's Bank Y. Problem: centralisation risks security & privacy.

Cryptocurrency = decentralised digital currency

- No central bank/government control – rules set by the crypto community
- Uses cryptography to track transactions
- Transactions are public and trackable
- Runs on a blockchain network – much more secure

Blockchain – How It Works

Each block contains: transaction data, a unique hash value (its "fingerprint"), and the previous block's hash (linking the chain).

	Block 1	Block 2	Block 3	Block 4
hash:	A4BF	6AB1	34EE	FF12
prev-hash:	0000	A4BF	6AB1	34EE

Fig 5.9 – Block 1 is the "genesis block" (no previous hash)

Exam tip – if a hacker changes data in Block 2, its hash changes, breaking the chain link to Block 3 onward, so tampering is instantly detectable. "Proof-of-work" (policed by miners) slows down block creation so hackers can't out-race the network.

Blockchain uses: cryptocurrency exchanges, smart contracts, research, politics, education.

5.3 Cyber Security - Threats

8 threats: Brute force - Data interception - DDoS - Hacking - Malware - Phishing - Pharming - Social engineering

Brute Force Attack

Systematically tries every combination until the password is cracked. Attackers often start with common passwords (123456, password, qwerty...) then a word list before full trial-and-error. Longer + more varied password = harder to crack.

Data Interception

- Wired networks – packet sniffer examines data packets
- Wireless – wardriving (laptop + antenna + GPS to catch Wi-Fi signals)
- Defence: encryption + WEP protocol + firewall; avoid public Wi-Fi for sensitive data

DDoS (Distributed Denial of Service)

Floods a server with useless traffic from many different computers so real users can't get through. Signs: slow network, can't access sites, spam flood.

Hacking

Illegal access to a system without permission – can lead to identity theft, data deletion/corruption. Prevented by firewalls, strong/changing passwords. Ethical hacking = companies pay hackers to test their own security legally.

Malware - The 6 Types

Type	*Key feature*
Virus	Replicates; needs an active host program to run and cause harm
Worm	Standalone, self-replicates without a host - spreads across whole networks
Trojan horse	Disguised as legitimate software; needs the user to run it
Spyware	Secretly monitors user activity (incl. keylogging), sends data back to attacker
Adware	Floods user with unwanted ads/pop-ups; hijacks browser
Ransomware	Encrypts the user's data, demands payment for the decryption key

> *Exam trap* - Virus NEEDS a host program to trigger it. Worm does NOT need a host and self-spreads - that's the key distinguishing exam point.

Phishing vs Pharming

Phishing	*Pharming*
Fake-looking legit emails trick user into clicking a link	Malicious code redirects the browser to a fake site
User MUST take an action (click) for harm to occur	User does NOT need to do anything - happens automatically
Prevention: check links, https padlock, anti-phishing toolbar	Caused by DNS cache poisoning; prevention: check URL spelling, anti-virus

Social Engineering

Tricks people (not systems) into breaking security. Exploits: fear (fake virus pop-up), curiosity (infected USB "baiting"), trust (fake "IT support" phone call).

5.3.2 Keeping Data Safe - Solutions

Access Levels & Authentication

Different users get different rights (e.g. a hospital cleaner vs a consultant). Social media typically has 4 levels: public – friends – custom – owner-only.

3 authentication factors: something you know (password/PIN) – something you have (phone) – something you are (biometrics).

Strong Passwords

> *Strong:* Sy12@#TT90kj=0 (capital + number + symbol)
> *Weak:* GREEN, a pet's name, a birthdate

Biometrics Comparison

Method	Accuracy	Drawback
Fingerprint	~1 in 5000	Fails if fingers dirty/damaged
Retina scan	~1 in 10 million	Slow, intrusive, expensive
Face recognition	Lower	Affected by lighting, age, glasses
Voice recognition	Lowest	Illness/recording can fool it

Firewalls vs Proxy Servers

Firewall	Proxy Server
Filters traffic in/out based on criteria	Sits between user & web server, forwards requests
Blocks/warns on suspicious traffic	Hides the user's real IP address
Can be software or hardware	Can cache pages for faster repeat access

SSL - Secure Sockets Layer

Protocol that encrypts data between browser and web server. Shown by https + padlock. Used in: online banking, shopping, email, cloud storage.

Past-Paper Style Practice

Cambridge-style questions written for self-practice, with a model marking scheme below each.

Q1

[3]

Explain the difference between a session cookie and a persistent cookie.

Mark scheme (max 3):

- Session cookie has no expiry date / stored in temporary memory
- Persistent cookie has an expiry date / stored on the hard drive
- Session cookie deleted when browser closes; persistent remains until expiry/deletion

Q2

[4]

Describe the difference between phishing and pharming.

Mark scheme (max 4):

- Phishing uses fake-looking emails to trick the user into clicking a link
- Requires the user to take an action for the attack to succeed
- Pharming uses malicious code to redirect the browser to a fake site
- Requires no action from the user (can happen via DNS cache poisoning)

Q3

[2]

State two features that distinguish a worm from a virus.

Mark scheme (max 2):

- A worm is standalone and does not need a host program; a virus does
- A worm self-replicates across a network without user action; a virus needs to be triggered

Q4

[3]

Explain how blockchain prevents a transaction record from being tampered with.

Mark scheme (max 3):

- Each block contains a hash of its own data and the previous block's hash
- Changing data in a block changes its hash, breaking the link to the next block
- Proof-of-work slows down block creation, making it impractical to fake a new chain quickly

Q5

[2]

Give two ways a firewall helps protect a computer from cyber threats.

Mark scheme (max 2):

- Examines/filters incoming and outgoing traffic against set criteria
- Blocks suspicious traffic and warns the user / logs traffic for review

Before your exam – Redraw the DNS lookup diagram and the blockchain diagram from memory. Practice writing out the malware table without looking – examiners love asking "identify this malware from the description."